Erasmus University Rotterdam
Erasmus School of Economics
Master's Thesis Econometrics and Management Science - Business Analytics and Quantitative Marketing

Median-of-Means Two-Stage Least Squares Regression

David von Schledorn Márquez (597334)

Supervisor:	Mikhail Zhelonkin
Second assessor:	-
Date final version:	-

Abstract

Median-of-Means methods are used for instrumental variable regression and their properties are compared to the standard 2-stage least squares method.

1 Preliminaries

1.1 Setup and Assumptions

Let $(Y_i, X_i, Z_i)_{i=1}^n$ be an i.i.d. random sample from a joint distribution on $\mathbb{R}^3$ satisfying the structural equation

$$Y_i = \beta X_i + \varepsilon_i, \tag{1}$$

where Y_i is the variable of interest, X_i is an endogenous regressor, and Z_i is the instrumental variable (IV). Additionally, $\beta \in \mathbb{R}$ is the structural parameter of interest and ε_i is an unobserved error term.

For any random quantity f, we write $\mu_f := \mathbb{E}[f]$ and $\sigma_f^2 := \mathrm{Var}(f)$ for its mean and variance. All population quantities below are instances of this scheme; for example, $\mu_X = \mathbb{E}[X_i]$, $\mu_{ZX} = \mathbb{E}[Z_i X_i]$, and $\sigma_{ZX}^2 = \mathrm{Var}(Z_i X_i)$.

MoM can also be in sub-script

We impose the following conditions.

(A1) **Exogeneity.** $\mu_{Z\varepsilon} = 0$.

(A2) **Relevance.** $\mu_{ZX} \neq 0$.

(A3) **Finite second moments.** $\sigma_{ZX}^2,\ \sigma_{ZY}^2,\ \sigma_{Z\varepsilon}^2 < \infty$.

By (A1) and (A2), the structural parameter is identified as $\beta = \mu_{ZY}/\mu_{ZX}$.

Remark 1.1 (Moment conditions)**.** *The conditions (A1)–(A3) hold for any distribution with finite second moments of the products Z_iY_i, Z_iX_i, and $Z_i\varepsilon_i$. Thus, no assumptions are imposed on the marginal distributions of Y_i, X_i, Z_i, or ε_i.*

Remark 1.2 (Relationship between variances)**.** *The three variances in (A3) are related. Since $Z_iY_i = \beta Z_iX_i + Z_i\varepsilon_i$, Minkowski's inequality gives*

$$\sigma_{Z\varepsilon} \leq \sigma_{ZY} + |\beta|\,\sigma_{ZX}. \tag{2}$$

To see this, set $U = Z_iY_i$ and $V = \beta Z_iX_i$, so that $U - V = Z_i\varepsilon_i$. By Minkowski's inequality for $p = 2$,

$$\sqrt{\mathrm{Var}(U - V)} \leq \sqrt{\mathrm{Var}(U)} + \sqrt{\mathrm{Var}(V)},$$

which gives $\sigma_{Z\varepsilon} \leq \sigma_{ZY} + |\beta|\,\sigma_{ZX}$.

1.2 Sub-Gaussian Estimators

Definition 1.3 (Sub-Gaussian Estimators)**.** *Let $(X_i)_{i=1}^n$ be a random sample with mean μ_X and variance σ_X^2, and let $\hat{\mu}_X$ denote an estimator for μ_X. For some constant $L > 0$, $\hat{\mu}_X$ is said to* *be a sub-Gaussian estimator for μ_X if*

$$\mathbb{P}\left[|\hat{\mu}_X - \mu_X| > L\sigma_X\sqrt{\frac{\ln(1/\delta)}{n}}\right] \leq \delta \quad \textit{for all } \delta \in (0,1). \tag{3}$$

1.3 The Median-of-Means Estimator for a Scalar Mean

We present the scalar Median-of-Means (MoM) estimator and its concentration properties. The results in this section are adapted from Devroye, Lerasle, Lugosi and Oliveira (2016) and Lugosi and Mendelson (2019).

Algorithm 1: Scalar Median-of-Means Estimator

Input: Sample $(X_i)_{i=1}^n$, number of blocks k

Output: Estimate $\hat{\mu}_{\mathrm{MoM}}$

Partition $(X_i)_{i=1}^n$ into k blocks $B_1, \ldots, B_k$ of size $m = \lfloor n/k \rfloor$;

for $j = 1, \ldots, k$ **do**

  Compute the block mean $\hat{\mu}_X^{(j)} = \frac{1}{m}\sum_{i \in B_j} X_i$;

return $\hat{\mu}_{\mathrm{MoM}} = \mathrm{med}(\hat{\mu}_X^{(1)}, \ldots, \hat{\mu}_X^{(k)})$;

Think about μ notation

Theorem 1.4 (MoM concentration)**.** *Let $(X_i)_{i=1}^n$ be i.i.d. with mean μ_X and variance $\sigma_X^2 < \infty$. Let k be a positive integer, $m = \lfloor n/k \rfloor$, and $\hat{\mu}_{\mathrm{MoM}}$ the MoM estimator of μ_X. Then*

$$\mathbb{P}\left[|\hat{\mu}_{\mathrm{MoM}} - \mu_X| > \frac{2\sigma_X}{\sqrt{m}}\right] \leq e^{-k/8}. \tag{4}$$

Setting $k = \lceil 8\ln(1/\delta) \rceil$ for $\delta \in (0,1)$, this becomes

$$\mathbb{P}\left[|\hat{\mu}_{\mathrm{MoM}} - \mu_X| > \sigma_X\sqrt{\frac{32\ln(1/\delta)}{n}}\right] \leq \delta, \tag{5}$$

which is sub-Gaussian in the sense of definition 1.3 with $L = \sqrt{32}$.

Proof. The proof proceeds in three steps.

Step 1 (Chebyshev on each block). Each block mean satisfies $\mathbb{E}[\bar{X}^{(j)}] = \mu_X$ and $\mathrm{Var}(\bar{X}^{(j)}) = \sigma_X^2/m$. By Chebyshev's inequality with threshold $2\sigma_X/\sqrt{m}$,

$$\mathbb{P}\left[\left|\bar{X}^{(j)} - \mu_X\right| > \frac{2\sigma_X}{\sqrt{m}}\right] \leq \frac{\sigma_X^2/m}{4\sigma_X^2/m} = \frac{1}{4}.$$

Step 2 (Counting argument for the median). Define $\zeta^{(j)} = \mathbf{1}\{|\bar{X}^{(j)} - \mu_X| > 2\sigma_X/\sqrt{m}\}$. By Step 1, $\mathbb{E}[\zeta^{(j)}] \leq 1/4$. Since the blocks are disjoint subsets of an i.i.d. sample, the $(\zeta^{(j)})_{j=1}^k$ are independent.

If $|\hat{\mu}_{\mathrm{MoM}} - \mu_X| > 2\sigma_X/\sqrt{m}$, the median exceeds $\mu_X + 2\sigma_X/\sqrt{m}$ or falls below $\mu_X - 2\sigma_X/\sqrt{m}$. In either case, at least $k/2$ of the block means satisfy $|\bar{X}^{(j)} - \mu_X| > 2\sigma_X/\sqrt{m}$, since the median of k numbers can only lie outside an interval if fewer than half the numbers lie inside it. Therefore,

$$\mathbb{P}\left[|\hat{\mu}_{\mathrm{MoM}} - \mu_X| > \frac{2\sigma_X}{\sqrt{m}}\right] \leq \mathbb{P}\left[\sum_{j=1}^k \zeta^{(j)} \geq \frac{k}{2}\right].$$

Step 3 (Hoeffding's inequality). The sum $\sum_{j=1}^{k} \zeta^{(j)}$ consists of k independent random variables with $\zeta^{(j)} \in [0,1]$ and $\mathbb{E}[\sum_j \zeta^{(j)}] \leq k/4$. For $\sum_j \zeta^{(j)}$ to reach $k/2$, it must exceed its mean by at least $k/2 - k/4 = k/4$. By Hoeffding's inequality (Hoeffding, 1963),

$$\mathbb{P}\left[\sum_{j=1}^{k} \zeta^{(j)} - \mathbb{E}\left[\sum_j \zeta^{(j)}\right] \geq \frac{k}{4}\right] \leq \exp\left(\frac{-2(k/4)^2}{k}\right) = e^{-k/8}.$$

This establishes (4). The sub-Gaussian form (5) follows by setting $k = \lceil 8 \ln(1/\delta) \rceil$. $\square$

2 The Standard Instrumental Variable Estimator

Definition 2.1 (Standard IV estimator). *Consider the structural equation* (1). *The standard IV estimator* $\hat{\beta}_{\mathrm{IV}}$ *is defined by*

$$\hat{\beta}_{\mathrm{IV}} = \frac{\hat{\mu}_{ZY}}{\hat{\mu}_{ZX}},$$

Lemma 2.2 (Error decomposition). *Under* (1) *and (A1),*

$$\hat{\beta}_{\mathrm{IV}} - \beta = \frac{\hat{\mu}_{Z\varepsilon}}{\hat{\mu}_{ZX}}, \tag{6}$$

where $\mathbb{E}[\hat{\mu}_{Z\varepsilon}] = 0$.

Proof. Multiply both sides of $Y_i = \beta X_i + \varepsilon_i$ by Z_i and average:

$$\hat{\mu}_{ZY} = \frac{1}{n} \sum_{i=1}^{n} Z_i(\beta X_i + \varepsilon_i) = \beta \hat{\mu}_{ZX} + \hat{\mu}_{Z\varepsilon}.$$

Rearranging and dividing by $\hat{\mu}_{ZX}$ gives $\hat{\beta}_{\mathrm{IV}} = \beta + \hat{\mu}_{Z\varepsilon}/\hat{\mu}_{ZX}$, which is (6). By (A1), $\mathbb{E}[Z_i \varepsilon_i] = 0$, so $\mathbb{E}[\hat{\mu}_{Z\varepsilon}] = 0$ by linearity of expectation. $\square$

Theorem 2.3 (Standard IV concentration). *Assume (A1)–(A3). For any* $\delta \in (0,1)$, *if*

$$n \geq \frac{8\sigma_{ZX}^2}{\delta \mu_{ZX}^2}, \tag{7}$$

then

$$\mathbb{P}\left[\left|\hat{\beta}_{\mathrm{IV}} - \beta\right| > \frac{2\sqrt{2}\sigma_{Z\varepsilon}}{|\mu_{ZX}| \sqrt{\delta n}}\right] \leq \delta. \tag{8}$$

Proof. By Lemma 2.2, the estimation error equals the ratio $\hat{\mu}_{Z\varepsilon}/\hat{\mu}_{ZX}$. To bound this ratio we need two things simultaneously: the denominator must be bounded away from zero, and the numerator must be small. We control each requirement with a separate probability event and then combine them via the union bound.

Step 1 (Two-event decomposition). For a denominator threshold $D \in (0, |\mu_{ZX}|)$ and an error threshold $t > 0$, both to be chosen below, define

$$A(D) = \{|\hat{\mu}_{ZX}| \geq D\}, \qquad B(D,t) = \{|\hat{\mu}_{Z\varepsilon}| \leq tD\}.$$

On $A(D) \cap B(D,t)$, the denominator satisfies $|\hat{\mu}_{ZX}| \geq D$ by definition of $A(D)$, so

$$\left|\hat{\beta}_{\mathrm{IV}} - \beta\right| = \frac{|\hat{\mu}_{Z\varepsilon}|}{|\hat{\mu}_{ZX}|} \leq \frac{tD}{D} = t.$$

Hence $\mathbb{P}\left[\left|\hat{\beta}_{IV} - \beta\right| > t\right] \leq \mathbb{P}[A(D)^c] + \mathbb{P}[B(D,t)^c]$ by the union bound. For the remainder of this proof we set $D = |\mu_{ZX}|/2$, the symmetric choice that yields the cleanest constants.

Step 2 (Chebyshev bounds). We bound each failure probability in turn.

The failure event $A^c = \{|\hat{\mu}_{ZX}| < |\mu_{ZX}|/2\}$ implies a large deviation of $\hat{\mu}_{ZX}$ from its mean. Specifically, if $|\hat{\mu}_{ZX}| < |\mu_{ZX}|/2$ then, by the reverse triangle inequality,

$$|\hat{\mu}_{ZX} - \mu_{ZX}| \geq |\mu_{ZX}| - |\hat{\mu}_{ZX}| > |\mu_{ZX}| - \frac{|\mu_{ZX}|}{2} = \frac{|\mu_{ZX}|}{2}.$$

Hence $A^c \subseteq \{|\hat{\mu}_{ZX} - \mu_{ZX}| > |\mu_{ZX}|/2\}$. Since $\mathrm{Var}(\hat{\mu}_{ZX}) = \sigma^2_{ZX}/n$, Chebyshev's inequality applied with threshold $|\mu_{ZX}|/2$ gives

$$\mathbb{P}[A^c] \leq \frac{\sigma^2_{ZX}/n}{\mu^2_{ZX}/4} = \frac{4\sigma^2_{ZX}}{n\mu^2_{ZX}}.$$

The failure event is $B^c = \{|\hat{\mu}_{Z\varepsilon}| > t\,|\mu_{ZX}|/2\}$. Since $\mathbb{E}[\hat{\mu}_{Z\varepsilon}] = 0$ and $\mathrm{Var}(\hat{\mu}_{Z\varepsilon}) = \sigma^2_{Z\varepsilon}/n$, Chebyshev's inequality gives

$$\mathbb{P}[B^c] \leq \frac{\sigma^2_{Z\varepsilon}/n}{t^2\mu^2_{ZX}/4} = \frac{4\sigma^2_{Z\varepsilon}}{nt^2\mu^2_{ZX}}.$$

Step 3 (Allocate the error budget and solve for t). We require the total failure probability $\mathbb{P}[A^c] + \mathbb{P}[B^c]$ to be at most δ, and we allocate half the budget to each event.

Budget for A^c. Setting $\mathbb{P}[A^c] \leq \delta/2$ requires

$$\frac{4\sigma^2_{ZX}}{n\mu^2_{ZX}} \leq \frac{\delta}{2} \iff n \geq \frac{8\sigma^2_{ZX}}{\delta\mu^2_{ZX}},$$

which is exactly condition (7).

Setting $\mathbb{P}[B^c] \leq \delta/2$ and solving for t:

$$\frac{4\sigma^2_{Z\varepsilon}}{nt^2\mu^2_{ZX}} = \frac{\delta}{2} \implies t = \frac{2\sqrt{2}\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{\delta n}}.$$

By the union bound,

$$\mathbb{P}\left[\left|\hat{\beta}_{\mathrm{IV}} - \beta\right| > t\right] \leq \mathbb{P}[A^c] + \mathbb{P}[B^c] \leq \frac{\delta}{2} + \frac{\delta}{2} = \delta. \qquad \square$$

Remark 2.4 (Polynomial dependence on δ)**.** *The bound* (8) *contains the factor $1/\sqrt{\delta}$, which is polynomial in $1/\delta$. Devroye et al. (2016) show that this is unavoidable: for any estimator formed from the empirical mean of i.i.d. observations with only finite variance, the $1/\sqrt{\delta}$ rate cannot be improved without additional distributional assumptions. The MoM-based estimators in Sections 3–4 circumvent this by operating on block means.*

Remark 2.5 (The symmetric threshold is not optimal)**.** *The choice $D = |\mu_{ZX}|/2$ was made for simplicity. It splits the distance between 0 and $|\mu_{ZX}|$ evenly, but this is not the split that minimises the error bound for given n and δ. Section 2.1 shows that treating D as a free parameter and optimising it reduces the instrument strength constant from 8 to 1 and improves the error constant by a factor of $2\sqrt{2}$.*

2.1 Optimised Threshold

The proof of Theorem 2.3 introduced a general threshold D but then fixed it symmetrically. Here we keep D free and choose it to minimise the resulting error bound, at the cost of a more involved Step 3.

Theorem 2.6 (Standard IV concentration, optimised threshold)**.** *Assume (A1)–(A3). Define*

$$\rho_{\mathrm{IV}} = \frac{\sigma_{ZX}^2}{\delta n \mu_{ZX}^2}.$$

For any $\delta \in (0,1)$, if $\rho_{\mathrm{IV}} < 1$ (i.e., $n > \sigma_{ZX}^2/(\delta\mu_{ZX}^2)$), then the choice $D^* = (1-\rho_{\mathrm{IV}}^{1/3})\,|\mu_{ZX}|$ is optimal and

$$\mathbb{P}\left[\left|\hat{\beta}_{\mathrm{IV}} - \beta\right| > \frac{\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{\delta n}(1-\rho_{\mathrm{IV}}^{1/3})^{3/2}}\right] \leq \delta. \tag{9}$$

Proof. Steps 1 and 2 are identical to those of Theorem 2.3 but with D left as a free parameter; we arrive at the Chebyshev bounds

$$\mathbb{P}[A(D)^c] \leq \frac{\sigma_{ZX}^2/n}{(|\mu_{ZX}|-D)^2}, \qquad \mathbb{P}[B(D,t)^c] \leq \frac{\sigma_{Z\varepsilon}^2/n}{t^2D^2}.$$

Requiring their sum to be at most δ gives the joint constraint

$$\frac{\sigma_{ZX}^2}{n(|\mu_{ZX}|-D)^2} + \frac{\sigma_{Z\varepsilon}^2}{nt^2D^2} \leq \delta. \tag{10}$$

Step 3 (Optimise D, then solve for t). Write $D = \eta\,|\mu_{ZX}|$ with $\eta \in (0,1)$. Dividing (10) through by δ and substituting $\rho_{\mathrm{IV}} = \sigma_{ZX}^2/(\delta n\mu_{ZX}^2)$:

$$\frac{\rho_{\mathrm{IV}}}{(1-\eta)^2} + \frac{\sigma_{Z\varepsilon}^2}{\delta n\mu_{ZX}^2\eta^2t^2} \leq 1.$$

Solving for t^2:

$$t^2 \geq \frac{\sigma_{Z\varepsilon}^2}{\delta n\mu_{ZX}^2} \cdot \frac{1}{f(\eta)}, \qquad f(\eta) = \eta^2\left(1 - \frac{\rho_{\mathrm{IV}}}{(1-\eta)^2}\right).$$

Minimising t is equivalent to maximising f over $\eta \in (0,1)$. Differentiating and setting $f'(\eta) = 0$:

$$f'(\eta) = 2\eta\left(1 - \frac{\rho_{\mathrm{IV}}}{(1-\eta)^2}\right) - \frac{2\rho_{\mathrm{IV}}\eta^2}{(1-\eta)^3} = 0.$$

Dividing through by $2\eta > 0$:

$$1 - \frac{\rho_{\mathrm{IV}}}{(1-\eta)^2} - \frac{\rho_{\mathrm{IV}}\eta}{(1-\eta)^3} = 0 \iff (1-\eta)^3 - \rho_{\mathrm{IV}}(1-\eta) - \rho_{\mathrm{IV}}\eta = 0 \iff (1-\eta)^3 = \rho_{\mathrm{IV}}.$$

Hence $\eta^* = 1 - \rho_{\mathrm{IV}}^{1/3}$, which lies in $(0,1)$ if and only if $\rho_{\mathrm{IV}} < 1$, the stated instrument strength condition. Substituting $(1-\eta^*)^2 = \rho_{\mathrm{IV}}^{2/3}$:

$$f(\eta^*) = (1-\rho_{\mathrm{IV}}^{1/3})^2\left(1 - \frac{\rho_{\mathrm{IV}}}{\rho_{\mathrm{IV}}^{2/3}}\right) = (1-\rho_{\mathrm{IV}}^{1/3})^2(1-\rho_{\mathrm{IV}}^{1/3}) = (1-\rho_{\mathrm{IV}}^{1/3})^3.$$

The minimised threshold is therefore

$$t^* = \frac{\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{\delta n}(1-\rho_{\mathrm{IV}}^{1/3})^{3/2}},$$

and the union bound gives $\mathbb{P}\left[\left|\hat{\beta}_{IV} - \beta\right| > t^*\right] \leq \delta$. □

Remark 2.7 (Comparison with the symmetric bound). *As $\rho_{\mathrm{IV}} \to 0$ (strong instruments or large n), the correction factor $(1-\rho_{\mathrm{IV}}^{1/3})^{-3/2} \to 1$ and (9) approaches $\sigma_{Z\varepsilon}/(|\mu_{ZX}|\sqrt{\delta n})$, a factor of $2\sqrt{2} \approx 2.83$ tighter than (8). As $\rho_{\mathrm{IV}} \to 1$ (approaching the boundary of instrument relevance), the factor diverges, reflecting the genuine difficulty of ratio estimation near instrument irrelevance. The instrument strength condition is eight times weaker than Theorem 2.3: $\rho_{\mathrm{IV}} < 1$ versus $\rho_{\mathrm{IV}} < 1/8$.*

2.2 Cantelli Improvement

The Chebyshev bound applied to $A(D)^c$ in the proofs above is two-sided. However, $A(D)^c = \{|\hat{\mu}_{ZX}| < D\}$ is a one-sided failure event: only a *downward* deviation of $\hat{\mu}_{ZX}$ from μ_{ZX} causes the denominator to be too small; an upward deviation makes it larger and is harmless. Cantelli's inequality exploits this one-sidedness.

Lemma 2.8 (Cantelli's inequality). *Let W be a random variable with $\mathbb{E}[W] = \mu$ and $\mathrm{Var}(W) = \sigma^2 < \infty$. For any $\tau > 0$,*

$$\mathbb{P}[W - \mu \leq -\tau] \leq \frac{\sigma^2}{\sigma^2 + \tau^2}.$$

This is strictly smaller than the Chebyshev bound σ^2/τ^2 for all $\tau > 0$, with the greatest relative gain when $\tau \approx \sigma$.

Theorem 2.9 (Standard IV concentration, Cantelli improvement). *Assume (A1)–(A3) and $\mu_{ZX} > 0$. Write $\gamma_{\mathrm{IV}} = \sigma_{ZX}^2/(n\mu_{ZX}^2)$ and define*

$$h(\eta) = \eta^2 \frac{\delta\big(\gamma_{\mathrm{IV}} + (1-\eta)^2\big) - \gamma_{\mathrm{IV}}}{\gamma_{\mathrm{IV}} + (1-\eta)^2}, \quad \eta \in (0,1). \tag{11}$$

If

$$n > \frac{(1-\delta)\sigma_{ZX}^2}{\delta\mu_{ZX}^2}, \quad \textit{equivalently,} \quad \gamma_{\mathrm{IV}} < \frac{\delta}{1-\delta}, \tag{12}$$

then $h(\eta) > 0$ on a non-empty interval and the estimator satisfies

$$\mathbb{P}\left[\left|\hat{\beta}_{\mathrm{IV}} - \beta\right| > \frac{\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{nh(\eta^*_{\mathrm{IV}})}}\right] \leq \delta, \tag{13}$$

where η^*_{IV} is the maximiser of h over its feasible domain.

Proof. We use the same two-event framework with $D = \eta\mu_{ZX}$ kept general.

Step 1 (Two-event decomposition). As in Theorem 2.3, define $A(D) = \{\hat{\mu}_{ZX} \geq D\}$ and $B(D,t) = \{|\hat{\mu}_{Z\varepsilon}| \leq tD\}$. On $A(D) \cap B(D,t)$, $\left|\hat{\beta}_{IV} - \beta\right| \leq t$.

Step 2 (Cantelli on A^c, Chebyshev on B^c).

Since $\mu_{ZX} > 0$ and $D < \mu_{ZX}$, the failure event satisfies

$$A(D)^c = \{\hat{\mu}_{ZX} < D\} \subseteq \{\hat{\mu}_{ZX} - \mu_{ZX} < -(\mu_{ZX} - D)\}.$$

This is a one-sided downward deviation, so Lemma 2.8 applies with $W = \hat{\mu}_{ZX}$, $\sigma^2 = \sigma^2_{ZX}/n$, and $\tau = \mu_{ZX} - D$:

$$\mathbb{P}[A(D)^c] \leq \frac{\sigma^2_{ZX}/n}{\sigma^2_{ZX}/n + (\mu_{ZX} - D)^2}.$$

The event $B(D,t)^c = \{|\hat{\mu}_{Z\varepsilon}| > tD\}$ is two-sided (a large residual of either sign causes failure), so Chebyshev is appropriate:

$$\mathbb{P}[B(D,t)^c] \leq \frac{\sigma^2_{Z\varepsilon}/n}{t^2D^2}.$$

Step 3 (Constraint and optimization). Setting the total failure probability to at most δ and substituting $D = \eta\mu_{ZX}$, $\gamma_{\mathrm{IV}} = \sigma^2_{ZX}/(n\mu^2_{ZX})$:

$$\frac{\gamma_{\mathrm{IV}}}{\gamma_{\mathrm{IV}} + (1-\eta)^2} + \frac{\sigma^2_{Z\varepsilon}}{nt^2\eta^2\mu^2_{ZX}} \leq \delta.$$

Isolating the second term and solving for t^2:

$$t^2 \geq \frac{\sigma^2_{Z\varepsilon}}{n\mu^2_{ZX}h(\eta)},$$

where $h(\eta)$ is defined in (11). The function $h(\eta)$ is positive only when $\delta(\gamma_{\mathrm{IV}} + (1-\eta)^2) > \gamma_{\mathrm{IV}}$, i.e., $(1-\eta)^2 > \gamma_{\mathrm{IV}}(1-\delta)/\delta$. The feasible region is non-empty if and only if $\gamma_{\mathrm{IV}} < \delta/(1-\delta)$, which is condition (12). Minimizing t over the feasible region by maximizing h and applying the union bound gives (13). □

Remark 2.10 (When Cantelli helps)**.** *Compared with the optimised Chebyshev condition ($n > \sigma^2_{ZX}/(\delta\mu^2_{ZX})$), the Cantelli condition* (12) *is weaker by a factor of* $1/(1-\delta)$. *At* $\delta = 0.05$ *this ratio is* $1/0.95 \approx 1.05$, *so the gain is negligible for typical significance levels. The improvement is more meaningful at moderate confidence levels: at* $\delta = 0.5$ *the factor reaches* 2. *For the standard IV estimator the Cantelli improvement is therefore primarily of theoretical interest. By contrast, for the MoR estimator (Section 4.2), Cantelli extends the valid range of the instrument strength parameter from* $\gamma < 1/4$ *to* $\gamma < 1/3$, *a qualitatively more substantial gain.*

Remark 2.11 (No closed form for η^*_{IV}). *The first-order condition $h'(\eta) = 0$ reduces to a quartic in η with no general closed-form solution. The optimised Chebyshev version (Theorem 2.6) yielded the clean form $\eta^* = 1 - \rho_{\mathrm{IV}}^{1/3}$ because the Chebyshev bound σ^2/τ^2 produces a simpler algebraic structure than the Cantelli bound $\sigma^2/(\sigma^2+\tau^2)$. The bound (13) must therefore be evaluated numerically for specific parameter values.*

3 The Ratio-of-Medians (RoM) Estimator

Algorithm 2: Ratio-of-Medians Estimator

Input: Sample $(Y_i, X_i, Z_i)_{i=1}^n$, number of blocks k

Output: Estimate $\hat{\beta}_{\mathrm{RM}}$

Partition $(Y_i, X_i, Z_i)_{i=1}^n$ into k blocks $B_1, \ldots, B_k$ of size $m = \lfloor n/k \rfloor$;

for $j = 1, \ldots, k$ **do**

  Compute the block means $\hat{\mu}^{(j)}_{ZY} = \frac{1}{m}\sum_{i \in B_j} Z_i Y_i$ and $\hat{\mu}^{(j)}_{ZX} = \frac{1}{m}\sum_{i \in B_j} Z_i X_i$;

Compute the coordinate-wise medians $\widetilde{\mu}_{ZY} = \mathrm{med}\left(\hat{\mu}^{(1)}_{ZY}, \ldots, \hat{\mu}^{(k)}_{ZY}\right)$ and $\widetilde{\mu}_{ZX} = \mathrm{med}\left(\hat{\mu}^{(1)}_{ZX}, \ldots, \hat{\mu}^{(k)}_{ZX}\right)$;

return $\hat{\beta}_{\mathrm{RM}} = \frac{\widetilde{\mu}_{ZY}}{\widetilde{\mu}_{ZX}}$;

geometric median

Theorem 3.1 (RoM concentration). *Assume (A1)–(A3). Let $\delta \in (0,1)$, $k = \lceil 8\ln(2/\delta) \rceil$, and $m = \lfloor n/k \rfloor$. If the instrument strength condition*

$$m > \frac{4\sigma^2_{ZX}}{\mu^2_{ZX}} \tag{14}$$

holds, then

$$\mathbb{P}\left[\left|\hat{\beta}_{\mathrm{RM}} - \beta\right| > \frac{2\left(\sigma_{ZY} + |\beta|\,\sigma_{ZX}\right)}{|\mu_{ZX}|\sqrt{m} - 2\sigma_{ZX}}\right] \le \delta. \tag{15}$$

Proof. The proof applies the scalar MoM result (Theorem 1.4) to each coordinate separately and combines the results with a union bound.

Step 1 (MoM on each coordinate). By Theorem 1.4 applied to the sequences $(Z_iY_i)_{i=1}^n$ and $(Z_iX_i)_{i=1}^n$,

$$\mathbb{P}\left[|\widetilde{\mu}_{ZY} - \mu_{ZY}| > \frac{2\sigma_{ZY}}{\sqrt{m}}\right] \le e^{-k/8}, \qquad \mathbb{P}\left[|\widetilde{\mu}_{ZX} - \mu_{ZX}| > \frac{2\sigma_{ZX}}{\sqrt{m}}\right] \le e^{-k/8}.$$

Step 2 (Union bound). Define the events

$$A = \left\{|\widetilde{\mu}_{ZY} - \mu_{ZY}| \le \frac{2\sigma_{ZY}}{\sqrt{m}}\right\}, \qquad B = \left\{|\widetilde{\mu}_{ZX} - \mu_{ZX}| \le \frac{2\sigma_{ZX}}{\sqrt{m}}\right\}.$$

By Step 1, $\mathbb{P}[A^c] \le e^{-k/8}$ and $\mathbb{P}[B^c] \le e^{-k/8}$. Note that A and B are not independent (both are constructed from the same data), so we use the union bound:

$$\mathbb{P}[(A \cap B)^c] = \mathbb{P}[A^c \cup B^c] \le \mathbb{P}[A^c] + \mathbb{P}[B^c] \le 2e^{-k/8}.$$

Setting $k = \lceil 8 \ln(2/\delta) \rceil$ ensures $2e^{-k/8} \le \delta$.

Step 3 (Algebraic bound on the ratio). On $A \cap B$, we decompose

$$\hat{\beta}_{\mathrm{RM}} - \beta = \frac{\widetilde{\mu}_{ZY}}{\widetilde{\mu}_{ZX}} - \frac{\mu_{ZY}}{\mu_{ZX}} = \frac{\widetilde{\mu}_{ZY}\mu_{ZX} - \mu_{ZY}\widetilde{\mu}_{ZX}}{\widetilde{\mu}_{ZX} \cdot \mu_{ZX}}.$$

Adding and subtracting $\mu_{ZY}\mu_{ZX}$ in the numerator and using $\mu_{ZY} = \beta\mu_{ZX}$,

$$\widetilde{\mu}_{ZY}\mu_{ZX} - \mu_{ZY}\widetilde{\mu}_{ZX} = \mu_{ZX}\left[(\widetilde{\mu}_{ZY} - \mu_{ZY}) - \beta(\widetilde{\mu}_{ZX} - \mu_{ZX})\right].$$

The factor μ_{ZX} cancels with the μ_{ZX} in the denominator, giving

$$\hat{\beta}_{\mathrm{RM}} - \beta = \frac{(\widetilde{\mu}_{ZY} - \mu_{ZY}) - \beta(\widetilde{\mu}_{ZX} - \mu_{ZX})}{\widetilde{\mu}_{ZX}}.$$

Taking absolute values and applying the triangle inequality to the numerator,

$$\left|\hat{\beta}_{\mathrm{RM}} - \beta\right| \le \frac{|\widetilde{\mu}_{ZY} - \mu_{ZY}| + |\beta|\,|\widetilde{\mu}_{ZX} - \mu_{ZX}|}{|\widetilde{\mu}_{ZX}|}.$$

Step 4 (Substitute bounds). On event A, $|\widetilde{\mu}_{ZY} - \mu_{ZY}| \le 2\sigma_{ZY}/\sqrt{m}$. On event B, $|\widetilde{\mu}_{ZX} - \mu_{ZX}| \le 2\sigma_{ZX}/\sqrt{m}$, and by the reverse triangle inequality,

$$|\widetilde{\mu}_{ZX}| \ge |\mu_{ZX}| - |\widetilde{\mu}_{ZX} - \mu_{ZX}| \ge |\mu_{ZX}| - \frac{2\sigma_{ZX}}{\sqrt{m}}.$$

The instrument strength condition (14) ensures $2\sigma_{ZX}/\sqrt{m} < |\mu_{ZX}|$, so the denominator is strictly positive. Substituting,

$$\left|\hat{\beta}_{\mathrm{RM}} - \beta\right| \le \frac{2\sigma_{ZY}/\sqrt{m} + |\beta| \cdot 2\sigma_{ZX}/\sqrt{m}}{|\mu_{ZX}| - 2\sigma_{ZX}/\sqrt{m}} = \frac{2(\sigma_{ZY} + |\beta|\,\sigma_{ZX})}{|\mu_{ZX}|\sqrt{m} - 2\sigma_{ZX}}.$$

Since this holds on $A \cap B$ and $\mathbb{P}[(A \cap B)^c] \le \delta$, the result follows. □

4 The Median-of-Ratios (MoR) Estimator

Algorithm 3: Median-of-Ratios Estimator

Input: Sample $(Y_i, X_i, Z_i)_{i=1}^n$, number of blocks k

Output: Estimate $\hat{\beta}_{\mathrm{MR}}$

Partition $(Y_i, X_i, Z_i)_{i=1}^n$ into k blocks $B_1, \dots, B_k$ of size $m = \lfloor n/k \rfloor$;

for $j = 1, \dots, k$ **do**

  Compute the block means $\hat{\mu}_{ZY}^{(j)} = \frac{1}{m}\sum_{i \in B_j} Z_i Y_i$ and $\hat{\mu}_{ZX}^{(j)} = \frac{1}{m}\sum_{i \in B_j} Z_i X_i$;

  Compute block level IV estimates $\hat{\beta}^{(j)} = \frac{\hat{\mu}_{ZY}^{(j)}}{\hat{\mu}_{ZX}^{(j)}}$;

return $\hat{\beta}_{\mathrm{MR}} = \mathrm{median}(\hat{\beta}^{(1)}, \dots, \hat{\beta}^{(k)})$;

Lemma 4.1 (Block-level error decomposition)**.** *Under* (1) *and (A1), for each block j,*

$$\hat{\beta}^{(j)} - \beta = \frac{\bar{\varepsilon}_j^Z}{\hat{\mu}_{ZX}^{(j)}}, \tag{16}$$

where $\bar{\varepsilon}_j^Z = \frac{1}{m}\sum_{i \in B_j} Z_i \varepsilon_i$ *satisfies* $\mathbb{E}[\bar{\varepsilon}_j^Z] = 0$.

Proof. Multiply $Y_i = \beta X_i + \varepsilon_i$ by Z_i and average over block B_j:

$$\hat{\mu}_{ZY}^{(j)} = \frac{1}{m}\sum_{i \in B_j} Z_i(\beta X_i + \varepsilon_i) = \beta \hat{\mu}_{ZX}^{(j)} + \bar{\varepsilon}_j^Z.$$

Dividing by $\hat{\mu}_{ZX}^{(j)}$ gives $\hat{\beta}^{(j)} = \beta + \bar{\varepsilon}_j^Z / \hat{\mu}_{ZX}^{(j)}$. By (A1) and linearity of expectation, $\mathbb{E}[\bar{\varepsilon}_j^Z] = \frac{1}{m}\sum_{i \in B_j} \mathbb{E}[Z_i \varepsilon_i] = 0$. □

Theorem 4.2 (MoR concentration)**.** *Assume (A1)–(A3). Let $\delta \in (0,1)$, $k = \lceil 8 \ln(1/\delta) \rceil$, and $m = \lfloor n/k \rfloor$. If the instrument strength condition*

$$m \geq \frac{32\sigma_{ZX}^2}{\mu_{ZX}^2} \tag{17}$$

holds, then

$$\mathbb{P}\left[\left|\hat{\beta}_{\mathrm{MR}} - \beta\right| > \frac{4\sqrt{2}\,\sigma_{Z\varepsilon}}{|\mu_{ZX}|\,\sqrt{m}}\right] \leq \delta. \tag{18}$$

Proof. The proof applies the MoM counting argument directly to the block-level IV estimates $\hat{\beta}^{(j)}$.

Step 1 (Two-event decomposition). By Lemma 4.1, $\left|\hat{\beta}^{(j)} - \beta\right| = \left|\bar{\varepsilon}_j^Z\right| / \left|\hat{\mu}_{ZX}^{(j)}\right|$. To bound this ratio, we control the numerator above and the denominator below. For a denominator threshold $D \in (0, |\mu_{ZX}|)$ and an error threshold $t > 0$, both to be chosen below, define

$$A_j(D) = \left\{\left|\hat{\mu}_{ZX}^{(j)}\right| \geq D\right\}, \qquad B_j(D,t) = \left\{\left|\bar{\varepsilon}_j^Z\right| \leq tD\right\}.$$

On $A_j(D) \cap B_j(D,t)$, the denominator satisfies $\left|\hat{\mu}_{ZX}^{(j)}\right| \geq D$ by definition of $A_j(D)$, so

$$\left|\hat{\beta}^{(j)} - \beta\right| = \frac{\left|\bar{\varepsilon}_j^Z\right|}{\left|\hat{\mu}_{ZX}^{(j)}\right|} \leq \frac{tD}{D} = t.$$

Hence $\mathbb{P}\left[\left|\hat{\beta}^{(j)} - \beta\right| > t\right] \leq \mathbb{P}[A_j(D)^c] + \mathbb{P}[B_j(D,t)^c]$ by the union bound. For the remainder of this proof we set $D = |\mu_{ZX}|/2$, the symmetric choice that yields the cleanest constants.

Step 2 (Chebyshev bounds). The two failure events are the block-level analogues of those in the proof of Theorem 2.3, with $\hat{\mu}_{ZX}$ replaced by the block mean $\hat{\mu}_{ZX}^{(j)}$, $\hat{\mu}_{Z\varepsilon}$ by $\bar{\varepsilon}_j^Z$, and the full sample size n by the block size m. The same reverse-triangle-inequality and Chebyshev arguments therefore give

$$\mathbb{P}[A_j^c] \leq \frac{\sigma_{ZX}^2/m}{\mu_{ZX}^2/4} = \frac{4\sigma_{ZX}^2}{m\mu_{ZX}^2}, \qquad \mathbb{P}[B_j^c] \leq \frac{\sigma_{Z\varepsilon}^2/m}{t^2\mu_{ZX}^2/4} = \frac{4\sigma_{Z\varepsilon}^2}{mt^2\mu_{ZX}^2}.$$

Step 3 (Allocate the error budget and solve for t). Unlike in the standard IV proof, the target here is not δ but the per-block failure probability 1/4: this is the level at which the counting argument of Steps 4–5 boosts the median to the desired confidence. We require $\mathbb{P}[A_j^c] + \mathbb{P}[B_j^c] \leq 1/4$ and allocate half the budget to each event.

Budget for A_j^c. Setting $\mathbb{P}[A_j^c] \leq 1/8$ requires

$$\frac{4\sigma_{ZX}^2}{m\mu_{ZX}^2} \leq \frac{1}{8} \iff m \geq \frac{32\sigma_{ZX}^2}{\mu_{ZX}^2},$$

which is exactly condition (17).

Setting $\mathbb{P}[B_j^c] \leq 1/8$ and solving for t:

$$\frac{4\sigma_{Z\varepsilon}^2}{mt^2\mu_{ZX}^2} \leq \frac{1}{8} \implies t \geq \frac{\sqrt{32}\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{m}}.$$

By the union bound, each block estimate satisfies

$$\mathbb{P}\left[\left|\hat{\beta}^{(j)} - \beta\right| > t\right] \leq \mathbb{P}[A_j^c] + \mathbb{P}[B_j^c] \leq \frac{1}{8} + \frac{1}{8} = \frac{1}{4}.$$

Step 4 (Counting argument). Define $\zeta_j = \mathbf{1}\{|\hat{\beta}^{(j)} - \beta| > t\}$. By Step 3, $\mathbb{E}[\zeta_j] \leq 1/4$. Since the blocks are disjoint, the $(\zeta_j)_{j=1}^k$ are independent. If $|\hat{\beta}_{\mathrm{MR}} - \beta| > t$, the median of the block estimates lies outside $(\beta - t, \beta + t)$, which requires at least $k/2$ blocks to satisfy $|\hat{\beta}^{(j)} - \beta| > t$. Therefore,

$$\mathbb{P}\left[\left|\hat{\beta}_{\mathrm{MR}} - \beta\right| > t\right] \leq \mathbb{P}\left[\sum_{j=1}^k \zeta_j \geq \frac{k}{2}\right].$$

Step 5 (Hoeffding). By Hoeffding's inequality, with overshoot $k/2 - k/4 = k/4$,

$$\mathbb{P}\left[\sum_{j=1}^k \zeta_j \geq \frac{k}{2}\right] \leq \exp\left(\frac{-2(k/4)^2}{k}\right) = e^{-k/8}.$$

Setting $k = \lceil 8\ln(1/\delta) \rceil$ gives $e^{-k/8} \leq \delta$, completing the proof. □

Remark 4.3 (Logarithmic dependence on δ)**.** *Through $k = \lceil 8\ln(1/\delta) \rceil$, the bound* (18) *depends on $1/\delta$ only logarithmically. This is precisely the improvement over the standard IV estimator anticipated in Remark 2.4: the median-of-block-estimates construction replaces the polynomial factor $1/\sqrt{\delta}$ of* (8) *by $\sqrt{\ln(1/\delta)}$.*

Remark 4.4 (The symmetric threshold is not optimal)**.** *As in the standard IV estimator (Remark 2.5), the choice $D = |\mu_{ZX}|/2$ was made for simplicity rather than optimality. Section 4.1 treats D as a free parameter.*

4.1 Optimised Threshold

The proof of Theorem 4.2 introduced a general threshold D but then fixed it symmetrically. Here we keep D free and choose it to minimise the resulting error bound, at the cost of a more involved Step 3.

Theorem 4.5 (MoR concentration, optimised threshold)**.** *Assume (A1)–(A3). Let* $\delta \in (0,1)$, $k = \lceil 8\ln(1/\delta)\rceil$, *and* $m = \lfloor n/k \rfloor$. *Define*

$$\rho_{\mathrm{MR}} = \frac{4\sigma_{ZX}^2}{m\mu_{ZX}^2}.$$

If $\rho_{\mathrm{MR}} < 1$ *(i.e.,* $m > 4\sigma_{ZX}^2/\mu_{ZX}^2$*), then the choice* $D^* = (1-\rho_{\mathrm{MR}}^{1/3})\,|\mu_{ZX}|$ *is optimal and*

$$\mathbb{P}\left[\left|\hat{\beta}_{\mathrm{MR}} - \beta\right| > \frac{2\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{m}\,(1-\rho_{\mathrm{MR}}^{1/3})^{3/2}}\right] \le \delta. \tag{19}$$

Proof. Steps 1 and 2 are identical to those of Theorem 4.2 but with D left as a free parameter; we arrive at the Chebyshev bounds

$$\mathbb{P}[A_j(D)^c] \le \frac{\sigma_{ZX}^2/m}{(|\mu_{ZX}| - D)^2}, \qquad \mathbb{P}[B_j(D,t)^c] \le \frac{\sigma_{Z\varepsilon}^2/m}{t^2D^2}.$$

Requiring their sum to be at most $\frac{1}{4}$ gives the joint constraint

$$\frac{\sigma_{ZX}^2}{m(|\mu_{ZX}| - D)^2} + \frac{\sigma_{Z\varepsilon}^2}{mt^2D^2} \le \frac{1}{4}. \tag{20}$$

Step 3 (Optimise D, then solve for t). Writing $D = \eta\,|\mu_{ZX}|$ with $\eta \in (0,1)$ and multiplying (20) through by 4, the constraint becomes $\rho_{\mathrm{MR}}/(1-\eta)^2 + 4\sigma_{Z\varepsilon}^2/(m\mu_{ZX}^2\eta^2t^2) \le 1$. This is exactly the constraint optimised in the proof of Theorem 2.6, with ρ_{IV} replaced by ρ_{MR} and the prefactor $\sigma_{Z\varepsilon}^2/(\delta n\mu_{ZX}^2)$ by $4\sigma_{Z\varepsilon}^2/(m\mu_{ZX}^2)$. The same optimisation therefore yields $\eta^* = 1-\rho_{\mathrm{MR}}^{1/3}$, feasible if and only if $\rho_{\mathrm{MR}} < 1$, and

$$t^* = \frac{2\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{m}\,(1-\rho_{\mathrm{MR}}^{1/3})^{3/2}}.$$

At this threshold each block satisfies $\mathbb{P}[\left|\hat{\beta}^{(j)} - \beta\right| > t^*] \le 1/4$, and the counting argument and Hoeffding's inequality of Steps 4–5 in the proof of Theorem 4.2 then give $\mathbb{P}[\left|\hat{\beta}_{\mathrm{MR}} - \beta\right| > t^*] \le \delta$. □

4.2 Cantelli Improvement

As in the standard IV estimator (Section 2.2), the denominator failure event $A_j(D)^c$ is one-sided: only a downward deviation of $\hat{\mu}_{ZX}^{(j)}$ from μ_{ZX} makes the denominator too small, while an upward deviation is harmless. Replacing the two-sided Chebyshev bound on this event by Cantelli's inequality (Lemma 2.8) sharpens the instrument strength condition.

Theorem 4.6 (MoR concentration, Cantelli improvement)**.** *Assume (A1)–(A3) and* $\mu_{ZX} > 0$. *Write* $\gamma_{\mathrm{MR}} = \sigma_{ZX}^2/(m\mu_{ZX}^2)$ *and define*

$$h(\eta) = \frac{\eta^2((1-\eta)^2 - 3\gamma_{\mathrm{MR}})}{\gamma_{\mathrm{MR}} + (1-\eta)^2}, \quad \eta \in (0,1). \tag{21}$$

If

$$m > \frac{3\sigma_{ZX}^2}{\mu_{ZX}^2}, \quad \textit{equivalently,} \quad \gamma_{\mathrm{MR}} < \frac{1}{3}, \tag{22}$$

then $h(\eta) > 0$ *on a non-empty interval and the estimator satisfies*

$$\mathbb{P}\left[\left|\hat{\beta}_{\mathrm{MR}} - \beta\right| > \frac{2\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{mh(\eta_{\mathrm{MR}}^*)}}\right] \leq \delta, \tag{23}$$

where η_{MR}^* *is the maximiser of* h *over its feasible domain.*

Proof. We use the same two-event framework with $D = \eta\mu_{ZX}$ kept general.

Step 1 (Two-event decomposition). Since $\mu_{ZX} > 0$, we drop the absolute value on the denominator and define $A_j(D) = \{\hat{\mu}_{ZX}^{(j)} \geq D\}$ and $B_j(D,t) = \{\left|\bar{\varepsilon}_j^Z\right| \leq tD\}$. As in Theorem 4.2, $\left|\hat{\beta}^{(j)} - \beta\right| \leq t$ on $A_j(D) \cap B_j(D,t)$.

Step 2 (Cantelli on A_j^c, Chebyshev on B_j^c). The event $A_j(D)^c$ is a one-sided downward deviation, so Lemma 2.8 applies with $W = \hat{\mu}_{ZX}^{(j)}$, $\sigma^2 = \sigma_{ZX}^2/m$, and $\tau = \mu_{ZX} - D$. The event $B_j(D,t)^c = \{\left|\bar{\varepsilon}_j^Z\right| > tD\}$ is two-sided, so Chebyshev remains appropriate. As in the proof of Theorem 2.9, these give

$$\mathbb{P}[A_j(D)^c] \leq \frac{\sigma_{ZX}^2/m}{\sigma_{ZX}^2/m + (\mu_{ZX} - D)^2}, \qquad \mathbb{P}[B_j(D,t)^c] \leq \frac{\sigma_{Z\varepsilon}^2/m}{t^2D^2}.$$

Step 3 (Constraint and optimisation). Setting the per-block failure probability to at most 1/4 and substituting $D = \eta\mu_{ZX}$, $\gamma_{\mathrm{MR}} = \sigma_{ZX}^2/(m\mu_{ZX}^2)$:

$$\frac{\gamma_{\mathrm{MR}}}{\gamma_{\mathrm{MR}} + (1-\eta)^2} + \frac{\sigma_{Z\varepsilon}^2}{mt^2\eta^2\mu_{ZX}^2} \leq \frac{1}{4}.$$

Isolating the second term and solving for t^2:

$$t^2 \geq \frac{4\sigma_{Z\varepsilon}^2}{m\mu_{ZX}^2 h(\eta)},$$

where $h(\eta)$ is defined in (21). The function $h(\eta)$ is positive only when $\frac{1}{4}(\gamma_{\mathrm{MR}} + (1-\eta)^2) > \gamma_{\mathrm{MR}}$, i.e., $(1-\eta)^2 > 3\gamma_{\mathrm{MR}}$. The feasible region is non-empty if and only if $\gamma_{\mathrm{MR}} < \frac{1}{3}$, which is condition (22). Minimising t over the feasible region by maximising h and applying Hoeffding and the union bound gives (23). □

Remark 4.7 (No closed form for η_{MR}^*). *Exactly as for the standard IV estimator (Remark 2.11), the first-order condition* $h'(\eta) = 0$ *reduces to a quartic in* η *with no general closed-form solution, in contrast to the clean form* $\eta^* = 1 - \rho_{\mathrm{MR}}^{1/3}$ *of the optimised Chebyshev version (Theorem 4.5). The bound* (23) *must therefore be evaluated numerically.*

5 Inference

5.1 Test Inversion

Fix a null hypothesis $H_0\colon \beta = \beta_0$ and define

$$W_i(\beta_0) := Z_i\big(Y_i - \beta_0 X_i\big), \qquad i = 1, \dots, n. \tag{24}$$

Under H_0 we have $W_i(\beta_0) = Z_i\varepsilon_i$, so by (A1) $\mathbb{E}[W_i(\beta_0)] = \mathbb{E}[Z_i\varepsilon_i] = 0$ and by (A3) $\mathrm{Var}(W_i(\beta_0)) = \sigma^2_{Z\varepsilon} < \infty$.

Dufour (1997) explains why test inversion is necessary under weak instruments: in any model whose identification region

$$\Theta_0 := \{\theta \in \Theta : \mu_{ZX}(\theta) = 0\}$$

is non-empty, no confidence set procedure can be simultaneously valid and bounded with probability one. A Wald-type interval built around a point estimate is therefore not generally valid, whereas inverting a test of $H_0\colon \beta = \beta_0$ over a grid of null values remains so.

5.2 The Median-of-Means Anderson–Rubin Test

We partition the sample into k blocks as in the preceding sections and form, within each block B_j, the block mean of the moment (24). Because $W_i(\beta_0) = Z_iY_i - \beta_0 Z_iX_i$ is linear in β_0, the block mean is an *affine* function of β_0:

$$\bar{W}_j(\beta_0) := \frac{1}{m}\sum_{i\in B_j} W_i(\beta_0) = \hat{\mu}^{(j)}_{ZY} - \beta_0\,\hat{\mu}^{(j)}_{ZX}, \qquad j = 1, \dots, k. \tag{25}$$

The median-of-means AR statistic is the coordinate-wise median of these block means,

$$\widetilde{W}(\beta_0) := \mathrm{med}\big(\bar{W}_1(\beta_0), \dots, \bar{W}_k(\beta_0)\big). \tag{26}$$

Algorithm 4: Median-of-Means Anderson–Rubin Test

Input: Sample $(Y_i, X_i, Z_i)_{i=1}^n$, number of blocks k, null value β_0, threshold $\tau_n(\delta)$
Output: Decision: reject or fail to reject $H_0\colon \beta = \beta_0$
Partition $(Y_i, X_i, Z_i)_{i=1}^n$ into k blocks $B_1, \dots, B_k$ of size $m = \lfloor n/k \rfloor$;
for $j = 1, \dots, k$ **do**
  Compute $\hat{\mu}^{(j)}_{ZY} = \frac{1}{m}\sum_{i\in B_j} Z_iY_i$ and $\hat{\mu}^{(j)}_{ZX} = \frac{1}{m}\sum_{i\in B_j} Z_iX_i$;
  Form $\bar{W}_j(\beta_0) = \hat{\mu}^{(j)}_{ZY} - \beta_0\hat{\mu}^{(j)}_{ZX}$;
$\widetilde{W}(\beta_0) \leftarrow \mathrm{med}\big(\bar{W}_1(\beta_0), \dots, \bar{W}_k(\beta_0)\big)$;
return *reject if* $\left|\widetilde{W}(\beta_0)\right| > \tau_n(\delta)$*, else fail to reject*;

Theorem 5.1 (Size of the MoM-AR test)**.** *Assume (A1)–(A3). Fix $\delta \in (0,1)$, set $k = \lceil 8\ln(1/\delta)\rceil$ and $m = \lfloor n/k \rfloor$, and define*

$$\tau_n(\delta) := \sigma_{Z\varepsilon}\sqrt{\frac{32\ln(1/\delta)}{n}}. \tag{27}$$

Then under $H_0\colon \beta = \beta_0$,

$$\mathbb{P}_{H_0}\left[\,\left|\widetilde{W}(\beta_0)\right| > \tau_n(\delta)\,\right] \leq \delta. \tag{28}$$

Consequently the test of Algorithm 4 has size at most δ.

Proof. The proof reduces the statistic to the scalar median-of-means estimator of Section 1.3 and applies its concentration bound.

Step 1 (Moments of the block moment under H_0). Under $H_0\colon \beta = \beta_0$ we have $W_i(\beta_0) = Z_i\varepsilon_i$, so the $(W_i(\beta_0))_{i=1}^n$ are i.i.d. with $\mathbb{E}[W_i(\beta_0)] = \mathbb{E}[Z_i\varepsilon_i] = 0$ by (A1) and $\mathrm{Var}(W_i(\beta_0)) = \sigma^2_{Z\varepsilon} < \infty$ by (A3).

Step 2 (Reduction to the scalar MoM estimator). By (25) and (26), $\widetilde{W}(\beta_0)$ is the coordinate-wise median of the k block means of $(W_i(\beta_0))_{i=1}^n$; that is, it is exactly the scalar median-of-means estimator of Algorithm 1 applied to this sequence, estimating its mean $\mu = 0$. Theorem 1.4, taken with $\sigma^2 = \sigma^2_{Z\varepsilon}$ and $k = \lceil 8\ln(1/\delta)\rceil$, therefore gives

$$\mathbb{P}_{H_0}\left[\,\left|\widetilde{W}(\beta_0)\right| > \sigma_{Z\varepsilon}\sqrt{\tfrac{32\ln(1/\delta)}{n}}\,\right] \leq \delta.$$

The deviation on the left is precisely $\tau_n(\delta)$ of (27), so this is (28). □

5.3 The MoM-AR Confidence Set

Inverting the test of Theorem 5.1 produces a confidence set. Cite confidence set

Definition 5.2 (MoM-AR confidence set)**.** *For $\delta \in (0,1)$ and $\tau_n(\delta)$ as in (27), the MoM-AR confidence set is*

$$CS_{1-\delta} := \left\{\beta_0 \in \mathbb{R} : \left|\widetilde{W}(\beta_0)\right| \leq \tau_n(\delta)\right\}. \tag{29}$$

Theorem 5.3 (Coverage)**.** *Assume (A1)–(A3) and let $k = \lceil 8\ln(1/\delta)\rceil$. Then*

$$\mathbb{P}_\beta\left[\beta \in CS_{1-\delta}\right] \geq 1 - \delta.$$

Proof. The result is the test-inversion dual of the size bound. By Definition 5.2, the true parameter lies in the confidence set exactly when the statistic evaluated at the truth stays below the threshold:

$$\beta \in CS_{1-\delta} \iff \left|\widetilde{W}(\beta)\right| \leq \tau_n(\delta).$$

Evaluating the moment at the true parameter gives $W_i(\beta) = Z_i\varepsilon_i$, which has mean zero by (A1). Applying the size bound of Theorem 5.1 at the true null $\beta_0 = \beta$ therefore yields $\mathbb{P}_\beta\left[\,\left|\widetilde{W}(\beta)\right| > \tau_n(\delta)\,\right] \leq \delta$. Taking complements,

$$\mathbb{P}_\beta[\beta \in CS_{1-\delta}] = \mathbb{P}_\beta\left[\left|\widetilde{W}(\beta)\right| \leq \tau_n(\delta)\right] \geq 1 - \delta.$$ □

5.4 Geometry of the Confidence Set

Theorem 5.4 (Piecewise-affine structure)**.** *The map $\beta_0 \mapsto \widetilde{W}(\beta_0)$ is continuous and piecewise affine. There exist finitely many breakpoints $\beta^{(1)} < \beta^{(2)} < \cdots < \beta^{(M)}$, with $M \leq \binom{k}{2}$, such that* *on each open interval of* $\mathbb{R} \setminus \{\beta^{(1)}, \dots, \beta^{(M)}\}$ *the function* $\widetilde{W}$ *coincides with a single block mean* $\bar{W}_j$ *and is therefore affine.*

Proof. The median of k numbers is the value occupying rank $r := \lceil k/2 \rceil$ once the numbers are sorted. We therefore study how the sorted order of $\bar{W}_1(\beta_0), \dots, \bar{W}_k(\beta_0)$ varies with β_0.

Step 1 (Crossings are finite). For a pair $i \neq j$, the equation $\bar{W}_i(\beta_0) = \bar{W}_j(\beta_0)$ reads $\hat{\mu}_{ZY}^{(i)} - \beta_0 \hat{\mu}_{ZX}^{(i)} = \hat{\mu}_{ZY}^{(j)} - \beta_0 \hat{\mu}_{ZX}^{(j)}$, i.e. $(\hat{\mu}_{ZX}^{(j)} - \hat{\mu}_{ZX}^{(i)})\beta_0 = \hat{\mu}_{ZY}^{(j)} - \hat{\mu}_{ZY}^{(i)}$. If $\hat{\mu}_{ZX}^{(i)} \neq \hat{\mu}_{ZX}^{(j)}$ this has a unique solution; if the slopes are equal the lines are parallel and never cross (or coincide, in which case the pair may be merged). Hence the crossing set $\mathcal{C} := \{\beta_0 : \bar{W}_i(\beta_0) = \bar{W}_j(\beta_0) \text{ for some } i \neq j\}$ satisfies $|\mathcal{C}| \leq \binom{k}{2}$. Enumerate it as $\beta^{(1)} < \cdots < \beta^{(M)}$ and set $\beta^{(0)} := -\infty$, $\beta^{(M+1)} := +\infty$.

Step 2 (The order is constant between crossings). Fix an interval $I_\ell := (\beta^{(\ell)}, \beta^{(\ell+1)})$. For any pair $i \neq j$ the difference $D_{ij}(\beta_0) := \bar{W}_i(\beta_0) - \bar{W}_j(\beta_0)$ is affine, hence continuous, and vanishes only on $\mathcal{C}$. Since $I_\ell \cap \mathcal{C} = \varnothing$, D_{ij} has constant sign on I_ℓ. As this holds for every pair, the total ordering of $\bar{W}_1(\beta_0), \dots, \bar{W}_k(\beta_0)$ is constant on I_ℓ; let π_ℓ be the permutation realising it, so that $\bar{W}_{\pi_\ell(1)}(\beta_0) \leq \cdots \leq \bar{W}_{\pi_\ell(k)}(\beta_0)$ throughout I_ℓ.

Step 3 (The median is one block mean on each interval). By Step 2 the rank-r value on I_ℓ is always supplied by the same index, so $\widetilde{W}(\beta_0) = \bar{W}_{\pi_\ell(r)}(\beta_0)$ for all $\beta_0 \in I_\ell$; this is affine. With at most $M + 1 \leq \binom{k}{2} + 1$ intervals, $\widetilde{W}$ is piecewise affine.

Step 4 (Continuity). Let $i = \pi_{\ell-1}(r)$ and $j = \pi_\ell(r)$ be the active median indices on either side of a breakpoint $\beta^{(\ell)}$. If $i = j$ there is nothing to prove. If $i \neq j$, the rank-r slot changes occupant across $\beta^{(\ell)}$, which can occur only if lines i and j exchange ranks there; by continuity of D_{ij} this requires $\bar{W}_i(\beta^{(\ell)}) = \bar{W}_j(\beta^{(\ell)})$. Hence the left and right limits of $\widetilde{W}$ at $\beta^{(\ell)}$ coincide with this common value, and $\widetilde{W}$ is continuous. (If three or more lines meet at $\beta^{(\ell)}$, any reshuffling is confined to the indices sharing the common value there, and the same conclusion holds.) $\square$

Corollary 5.5 (Candidate boundary points)**.** *Every boundary point of* $CS_{1-\delta}$ *lies in the set*

$$\mathcal{B}_{\text{cand}} := \bigcup_{j:\, \hat{\mu}_{ZX}^{(j)} \neq 0} \left\{ \frac{\hat{\mu}_{ZY}^{(j)} - \tau_n(\delta)}{\hat{\mu}_{ZX}^{(j)}},\ \frac{\hat{\mu}_{ZY}^{(j)} + \tau_n(\delta)}{\hat{\mu}_{ZX}^{(j)}} \right\}, \tag{30}$$

a set of at most $2k$ *explicitly computable points.*

Proof. A boundary point β_0 of $CS_{1-\delta}$ lies on the boundary of the band, so $\left|\widetilde{W}(\beta_0)\right| = \tau_n(\delta)$, i.e. $\widetilde{W}(\beta_0) = +\tau_n(\delta)$ or $\widetilde{W}(\beta_0) = -\tau_n(\delta)$. By Theorem 5.4, β_0 lies in the closure of an interval on which $\widetilde{W}$ coincides with a single block mean $\bar{W}_j$, so it solves $\bar{W}_j(\beta_0) = \pm\tau_n(\delta)$ for some j. Two cases arise, according to the slope $-\hat{\mu}_{ZX}^{(j)}$ of that piece.

If $\hat{\mu}_{ZX}^{(j)} \neq 0$, then $\bar{W}_j$ is strictly monotone and $\hat{\mu}_{ZY}^{(j)} - \beta_0 \hat{\mu}_{ZX}^{(j)} = \pm\tau_n(\delta)$ has the unique solutions

$$\beta_0 = \frac{\hat{\mu}_{ZY}^{(j)} \mp \tau_n(\delta)}{\hat{\mu}_{ZX}^{(j)}},$$

namely the two values contributed by block j in (30). If $\hat{\mu}_{ZX}^{(j)} = 0$, then $\bar{W}_j \equiv \hat{\mu}_{ZY}^{(j)}$ is constant and equals $\pm\tau_n(\delta)$ only in the non-generic event $\hat{\mu}_{ZY}^{(j)} = \pm\tau_n(\delta)$; such a block contributes no isolated boundary point. Ranging over the k blocks, each of which contributes at most two candidates, gives $|\mathcal{B}_{\text{cand}}| \leq 2k$. □

Corollary 5.6 (Finite union of intervals). *$CS_{1-\delta}$ is a finite union of at most $k+1$ closed intervals, each possibly a single point or unbounded.*

Proof. By Theorem 5.4, $\left|\widetilde{W}\right|$ is continuous and piecewise affine. The set $CS_{1-\delta} = \{\beta_0 : \left|\widetilde{W}(\beta_0)\right| \leq \tau_n(\delta)\}$ is the preimage of the closed interval $[-\tau_n(\delta), \tau_n(\delta)]$ under this continuous function, hence closed; being a sublevel set of a piecewise-affine function with finitely many pieces, it is a finite union of closed intervals.

To count the components, note that the endpoints of these intervals are boundary points of $CS_{1-\delta}$, and by Corollary 5.5 there are at most $2k$ such points. Ordering them partitions $\mathbb{R}$ into at most $2k+1$ open intervals, on each of which $\left|\widetilde{W}\right| - \tau_n(\delta)$ has constant sign and so lies either entirely inside or entirely outside the band. Since the in-band and out-of-band intervals alternate along the line, at most $\lceil (2k+1)/2 \rceil = k+1$ of them lie in the band. Merging any in-band intervals that meet at a shared boundary point can only reduce this count, so $CS_{1-\delta}$ has at most $k+1$ components. □

5.5 Monotonicity and Single-Interval Confidence Sets

The slope of $\widetilde{W}$ on any affine piece is $-\hat{\mu}_{ZX}^{(j)}$ for the active index j. When all block slopes share a sign, $\widetilde{W}$ is monotonic and the confidence set is a single interval.

Proposition 5.7 (Deterministic single-interval condition). *If the block sums $\hat{\mu}_{ZX}^{(1)}, \ldots, \hat{\mu}_{ZX}^{(k)}$ all share the same sign, then $\widetilde{W}$ is monotonic on $\mathbb{R}$ and $CS_{1-\delta}$ is a single (possibly empty or unbounded) interval.*

Proof. By Theorem 5.4 the slope of $\widetilde{W}$ on each affine piece equals $-\hat{\mu}_{ZX}^{(j)}$ for the active index j. If all $\hat{\mu}_{ZX}^{(j)} > 0$ then every such slope is negative, so the continuous piecewise-affine function $\widetilde{W}$ is non-increasing on $\mathbb{R}$; if all $\hat{\mu}_{ZX}^{(j)} < 0$ it is non-decreasing. In either case $\widetilde{W}$ is monotone, so each of the level sets $\{\widetilde{W} \leq \tau_n(\delta)\}$ and $\{\widetilde{W} \geq -\tau_n(\delta)\}$ is a half-line (a ray). The confidence set $CS_{1-\delta}$ is their intersection, hence a single interval, which may be empty or unbounded. □

Proposition 5.8 (Single-interval condition, Chebyshev). *Assume (A1)–(A3) with $\mu_{ZX} \neq 0$. If*

$$m \geq \frac{k\,\sigma_{ZX}^2}{\delta\,\mu_{ZX}^2}, \tag{31}$$

then with probability at least $1-\delta$ all block sums $\hat{\mu}_{ZX}^{(j)}$ share the sign of μ_{ZX}; consequently $\widetilde{W}$ is monotone and $CS_{1-\delta}$ is a single interval.

Proof. Assume without loss of generality that $\mu_{ZX} > 0$; the case $\mu_{ZX} < 0$ is symmetric. By Proposition 5.7 it suffices to show that all block sums are positive with probability at least $1-\delta$, since sign-consistency then forces $\widetilde{W}$ to be monotone. Sign-consistency fails on the event

$$\mathcal{F} = \bigcup_{j=1}^{k} \{\hat{\mu}_{ZX}^{(j)} \leq 0\}.$$

Step 1 (Chebyshev on each block). A non-positive block sum is a large downward deviation from $\mu_{ZX} > 0$: if $\hat{\mu}_{ZX}^{(j)} \le 0$ then $\left|\hat{\mu}_{ZX}^{(j)} - \mu_{ZX}\right| = \mu_{ZX} - \hat{\mu}_{ZX}^{(j)} \ge \mu_{ZX}$, so $\{\hat{\mu}_{ZX}^{(j)} \le 0\} \subseteq \{\left|\hat{\mu}_{ZX}^{(j)} - \mu_{ZX}\right| \ge \mu_{ZX}\}$. Since $\mathrm{Var}(\hat{\mu}_{ZX}^{(j)}) = \sigma_{ZX}^2/m$, Chebyshev's inequality applied with threshold μ_{ZX} gives

$$\mathbb{P}\big[\hat{\mu}_{ZX}^{(j)} \le 0\big] \le \frac{\sigma_{ZX}^2/m}{\mu_{ZX}^2} = \frac{\sigma_{ZX}^2}{m\mu_{ZX}^2}.$$

Step 2 (Union bound over blocks). Summing over the k blocks,

$$\mathbb{P}[\mathcal{F}] \le \sum_{j=1}^{k} \mathbb{P}\big[\hat{\mu}_{ZX}^{(j)} \le 0\big] \le \frac{k\sigma_{ZX}^2}{m\mu_{ZX}^2},$$

which is at most δ precisely when $m \ge k\sigma_{ZX}^2/(\delta\mu_{ZX}^2)$, i.e. condition (31). On the complement $\mathcal{F}^c$, all block sums share the sign of μ_{ZX}, and Proposition 5.7 makes $CS_{1-\delta}$ a single interval. □

Remark 5.9 (Cantelli refinement)**.** *Replacing Chebyshev's inequality by the one-sided Cantelli bound (Lemma 2.8) in Step 1 sharpens the per-block probability, and hence the constant in condition* (31)*. As for the standard IV estimator (Remark 2.10), the resulting improvement is minor, so we retain the Chebyshev form.*

5.6 Computation

The confidence set can be computed exactly and in closed form. The idea is to enumerate the crossing points of the block-mean lines $\bar{W}_1, \ldots, \bar{W}_k$ to obtain the affine pieces of $\widetilde{W}$ (Theorem 5.4), identify the active median line on each piece, and retain the sub-intervals on which this line lies within the band $[-\tau_n(\delta), \tau_n(\delta)]$. Algorithm 5 collects these steps.

On each interval $(\beta^{(\ell)}, \beta^{(\ell+1)})$ the statistic $\widetilde{W}$ coincides with a single affine piece $\bar{W}_a$, so intersecting the band constraint $\left|\bar{W}_a(\beta_0)\right| \le \tau_n(\delta)$ with that interval is exact. By Corollary 5.6 the returned set consists of at most $k+1$ closed intervals, whose endpoints lie among the $2k$ candidates of Corollary 5.5.

Remark 5.10 (Feasibility and choice of variance estimator)**.** *In practice, $\sigma_{Z\varepsilon}$ is unknown and has to be estimated with a consistent estimator. A feasible test would replace this by a consistent estimator $\hat{\sigma}_{Z\varepsilon}$. This estimator must be computed independently of the null value β_0; test inversion compares the varying statistic $\widetilde{W}(\beta_0)$ against a single band across a grid of β_0 values. Forming residuals at the tested value is inconsistent whenever $\beta_0 \neq \beta$ since $Y_i - \beta_0 X_i = \varepsilon_i + (\beta - \beta_0)X_i$,*

$$\frac{1}{n}\sum_{i=1}^{n}\big(Z_i(Y_i - \beta_0 X_i)\big)^2 \xrightarrow{p} \sigma_{Z\varepsilon}^2 + 2(\beta - \beta_0)\,\mathbb{E}[Z^2 X\varepsilon] + (\beta - \beta_0)^2\,\mathbb{E}[(ZX)^2],$$

which grows as β_0 departs from β. This would lead to the band widening with the signal it is trying to detect, leading to a loss of power and a collapse of the constant-threshold geometry of Section 5.4.

A β_0-free estimate could be obtained instead from residuals at a preliminary consistent estimator. However, under the heavy tails permitted by (A3), $Z\varepsilon$ may have no finite fourth moment, in which case the estimator $\frac{1}{n}\sum_i (Z_i\hat{\varepsilon}_i)^2$ has infinite variance; the central limit theorem then

Algorithm 5: MoM-AR Confidence Set

Input: Sample $(Y_i, X_i, Z_i)_{i=1}^n$, number of blocks k, threshold $\tau_n(\delta)$
Output: Confidence set $CS_{1-\delta}$
Partition $(Y_i, X_i, Z_i)_{i=1}^n$ into k blocks $B_1, \dots, B_k$ of size $m = \lfloor n/k \rfloor$;
for $j = 1, \dots, k$ **do**
  Compute $\hat{\mu}_{ZY}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i Y_i$ and $\hat{\mu}_{ZX}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i X_i$;
$\mathcal{C} \leftarrow \varnothing$;
foreach *pair* $1 \le i < j \le k$ *with* $\hat{\mu}_{ZX}^{(i)} \neq \hat{\mu}_{ZX}^{(j)}$ **do**
  $\mathcal{C} \leftarrow \mathcal{C} \cup \{(\hat{\mu}_{ZY}^{(j)} - \hat{\mu}_{ZY}^{(i)})/(\hat{\mu}_{ZX}^{(j)} - \hat{\mu}_{ZX}^{(i)})\}$;
Sort $\mathcal{C}$ as $\beta^{(1)} < \cdots < \beta^{(M)}$ and set $\beta^{(0)} = -\infty$, $\beta^{(M+1)} = +\infty$;
$CS_{1-\delta} \leftarrow \varnothing$;
for $\ell = 0, \dots, M$ **do**
  Choose any $\beta_\star \in (\beta^{(\ell)}, \beta^{(\ell+1)})$;
  Find the active median index a with $\bar{W}_a(\beta_\star) = \operatorname{med}\big(\bar{W}_1(\beta_\star), \dots, \bar{W}_k(\beta_\star)\big)$;
  if $\hat{\mu}_{ZX}^{(a)} \neq 0$ **then**
    $J \leftarrow$ closed interval with endpoints $(\hat{\mu}_{ZY}^{(a)} \pm \tau_n(\delta))/\hat{\mu}_{ZX}^{(a)}$;
  else
    $J \leftarrow \mathbb{R}$ if $\left|\hat{\mu}_{ZY}^{(a)}\right| \le \tau_n(\delta)$, else $J \leftarrow \varnothing$;
  $CS_{1-\delta} \leftarrow CS_{1-\delta} \cup \big([\beta^{(\ell)}, \beta^{(\ell+1)}] \cap J\big)$;
return $CS_{1-\delta}$, *merging intervals that share an endpoint*;

no longer applies, so it converges slowly and behaves erratically. A robust scale estimator is therefore preferable.

The self-normalised variant discussed in Section 5.7 avoids the matter altogether, cancelling $\sigma_{Z\varepsilon}$ rather than estimating it.

5.7 Self-Normalised Anderson-Rubin Test

Definition 5.11 (Self-normalised statistic)**.** *For a vector $v = (v_1, \dots, v_k) \in \mathbb{R}^k$, let* $\operatorname{MAD}(v) = \operatorname{med}_j |v_j - \operatorname{med}_l v_l|$ *denote its median absolute deviation. With $\bar{W}_j$ and $\widetilde{W}$ as in* (25)–(26)*, the self-normalised statistic at β_0 is*

$$T(\beta_0) = \frac{\left|\widetilde{W}(\beta_0)\right|}{\hat{s}(\beta_0)}, \qquad \hat{s}(\beta_0) = \operatorname{MAD}\big(\bar{W}_1(\beta_0), \dots, \bar{W}_k(\beta_0)\big).$$

Given a critical value $c_{k,\delta}$, the test rejects $H_0\colon \beta = \beta_0$ when $T(\beta_0) > c_{k,\delta}$.

Proposition 5.12 (Scale invariance)**.** *Under H_0, the distribution of $T(\beta_0)$ does not depend on $\sigma_{Z\varepsilon}$. The same holds for any scale functional $\hat{s}$ that is homogeneous of degree one.*

Proof. Under H_0, $\bar{W}_j(\beta_0) = \frac{1}{m} \sum_{i \in B_j} Z_i \varepsilon_i = \sigma_{Z\varepsilon} G_j$, where $G_j = \frac{1}{m} \sum_{i \in B_j} Z_i \varepsilon_i / \sigma_{Z\varepsilon}$. The summands $Z_i \varepsilon_i / \sigma_{Z\varepsilon}$ have mean 0 and variance 1 by (A1) and (A3), so the joint distribution $(G_1, \dots, G_k)$ are free of $\sigma_{Z\varepsilon}$. Since med and MAD are homogeneous of degree one,

$$T(\beta_0) = \frac{|\operatorname{med}_j \sigma_{Z\varepsilon} G_j|}{\operatorname{MAD}_j(\sigma_{Z\varepsilon} G_j)} = \frac{\sigma_{Z\varepsilon}\, |\operatorname{med}_j G_j|}{\sigma_{Z\varepsilon} \operatorname{MAD}_j(G_j)} = \frac{|\operatorname{med}_j G_j|}{\operatorname{MAD}_j(G_j)},$$

and a function of $(G_1, \dots, G_k)$ alone; its distribution therefore does not involve $\sigma_{Z\varepsilon}$. The argument uses only $\mathrm{med}(\sigma v) = \sigma\,\mathrm{med}(v)$ and $\hat{s}(\sigma v) = \sigma \hat{s}(v)$, so it holds for any degree-one homogeneous $\hat{s}$. $\square$

Proposition 5.13 (Asymptotic null distribution)**.** *Assume (A1)–(A3), with k fixed and $m \to \infty$. Under H_0,*

$$T(\beta_0) \implies R_k := \frac{|\mathrm{med}(\xi_1, \dots, \xi_k)|}{\mathrm{MAD}(\xi_1, \dots, \xi_k)}, \qquad \xi_1, \dots, \xi_k \overset{iid}{\sim} N(0, 1),$$

and the distribution of R_k depends only on k.

Proof. Fix a block. By (A1) and (A3) the standardised summands $Z_i\varepsilon_i/\sigma_{Z\varepsilon}$ are i.i.d. with mean 0 and variance 1, so the central limit theorem gives $\sqrt{m}\,G_j = \frac{1}{\sqrt{m}} \sum_{i \in B_j} Z_i\varepsilon_i/\sigma_{Z\varepsilon} \Rightarrow \xi_j \sim N(0, 1)$. The blocks use disjoint observations, so $(\sqrt{m}\,G_1, \dots, \sqrt{m}\,G_k) \Rightarrow (\xi_1, \dots, \xi_k)$ with independent coordinates. Writing $T(\beta_0) = |\mathrm{med}_j \sqrt{m}\,G_j| \,/\, \mathrm{MAD}_j(\sqrt{m}\,G_j)$, the factor $\sqrt{m}$ cancels between numerator and denominator; as med and MAD are continuous, the continuous mapping theorem yields $T(\beta_0) \Rightarrow R_k$. The limit is a fixed functional of k standard normals, so its distribution depends only on k. $\square$

The critical value $c_{k,\delta}$ is the $(1-\delta)$ quantile of R_k; since R_k depends only on k, it is computed once by simulation and tabulated in Section 6.1

Corollary 5.14 (Confidence set)**.** *Inverting the self-normalised test gives*

$$CS_{1-\delta} = \left\{\beta_0 \in \mathbb{R} : \left|\widetilde{W}(\beta_0)\right| \le c_{k,\delta}\, \hat{s}(\beta_0)\right\},$$

a finite union of closed intervals.

Proof. By Theorem 5.4, $\widetilde{W}$ is continuous and piecewise affine; $\hat{s}(\beta_0) = \mathrm{MAD}_j(\bar{W}_j(\beta_0))$ is a median of absolute values of affine functions, hence also continuous and piecewise affine. On any interval where both coincide with single affine pieces $a(\beta_0)$ and $b(\beta_0) \ge 0$, the defining condition $-c_{k,\delta}\, b(\beta_0) \le a(\beta_0) \le c_{k,\delta}\, b(\beta_0)$ is a pair of linear inequalities, hence an interval. As there are finitely many pieces, $CS_{1-\delta}$ is a finite union of closed intervals. $\square$

Remark 5.15 (An exactly distribution-free variant)**.** *If $Z\varepsilon$ is symmetric about 0 under H_0, the signs of the block means are independent fair coin flips, so $N_+(\beta_0) = \#\{j : \bar{W}_j(\beta_0) > 0\} \sim$ $\mathrm{Binomial}(k, \frac{1}{2})$ exactly, free of both $\sigma_{Z\varepsilon}$ and the tail shape. Rejecting when $N_+(\beta_0)$ is far from $k/2$ gives a finite-sample-exact, distribution-free test, at the cost of the symmetry assumption and of lower power, since it uses only the signs of the block means.*

size and power

6 Appendix

6.1 Self-Normalised Anderson-Rubin Test table

fill out

References

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